Integrate $e^{\frac{-x^2}{2\sigma^2}}$

Show that for any $\sigma \neq 0$, $\quad \int_{-\infty}^{\infty} e^{-\frac{x^2}{2\sigma^2}}\,dx = \sqrt{2\pi}\,|\sigma|, \qquad \sigma \neq 0.$

Solution:

Step 1. Square the integral. Let

$$I = \int_{-\infty}^{\infty} e^{-\frac{x^2}{2\sigma^2}}\,dx.$$

Then

$$\begin{aligned} I^2 &= \left(\int_{-\infty}^{\infty} e^{-\frac{x^2}{2\sigma^2}}\,dx\right)\left(\int_{-\infty}^{\infty} e^{-\frac{y^2}{2\sigma^2}}\,dy\right) \\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-\frac{x^2+y^2}{2\sigma^2}}\,dxdy \end{aligned}$$

Step 2. Convert to polar coordinates. Use

$$x = r\cos\theta, \quad y = r\sin\theta, \quad x^2+y^2 = r^2, \quad dxdy = rdrd\theta.$$

Thus

$$I^2 = \int_0^{2\pi}\int_0^{\infty} e^{-\frac{r^2}{2\sigma^2}}\,rdrd\theta.$$

Step 3. Integrate with respect to r first. Consider the inner integral

$$\int_0^{\infty} re^{-\frac{r^2}{2\sigma^2}}\,dr.$$

Let

$$u = \frac{r^2}{2\sigma^2} \quad \Rightarrow \quad du = \frac{r}{\sigma^2}dr \quad \Rightarrow \quad r\,dr = \sigma^2\,du.$$

When $r = 0$, $u = 0$; when $r \to \infty$, $u \to \infty$. Hence

$$\int_0^{\infty} re^{-\frac{r^2}{2\sigma^2}}\,dr = \int_0^{\infty} \sigma^2 e^{-u}du = \sigma^2\int_0^{\infty} e^{-u}du = \sigma^2.$$

Substitute this back into the double integral:

$$I^2 = \int_0^{2\pi}\left(\int_0^{\infty} re^{-\frac{r^2}{2\sigma^2}}\,dr\right)d\theta = \int_0^{2\pi}\sigma^2\,d\theta.$$

Step 4. Integrate with respect to θ.

$$I^2 = \sigma^2\int_0^{2\pi} d\theta = \sigma^2 \cdot 2\pi = 2\pi\sigma^2.$$

Step 5. Take the square root.

$$I^2 = 2\pi\sigma^2 \quad \Rightarrow \quad I = \sqrt{2\pi\sigma^2} = \sqrt{2\pi}\,|\sigma|.$$

Therefore,

$$\int_{-\infty}^{\infty} e^{-\frac{x^2}{2\sigma^2}}\,dx = \sqrt{2\pi}|\sigma|, \quad \text{where } \sigma \neq 0.$$